Java Abstract Class & Adapter Class Notes

1. What is an Abstract Class?

An abstract class in Java is a class that cannot be instantiated directly. It is used as a blueprint for other classes. Abstract classes can contain: Abstract methods (without body) Concrete methods (with body) Variables and constructors

Exam Definition: An abstract class is a class declared with the abstract keyword that may contain abstract and non-abstract methods.

2. Why Use Abstract Classes?

- To achieve abstraction
- To provide common functionality
- To reduce code duplication
- To create a parent blueprint for child classes

3. Features of Abstract Class

Feature	Description
Cannot create object	Abstract classes cannot be instantiated
Can contain abstract methods	Methods without body
Can contain concrete methods	Methods with implementation
Supports inheritance	Child classes extend abstract class
Can have constructors	Used during object creation

4. Abstract Method

An abstract method is a method without a body. The child class must override and implement it.

```
abstract class Animal {
    abstract void sound();
}
```

5. Abstract Class Example

```
abstract class Animal {

    abstract void sound();

    void sleep() {
        System.out.println("Sleeping...");
    }
}

class Dog extends Animal {

    void sound() {
```

```
        System.out.println("Bark");
    }
}

public class Main {
    public static void main(String[] args) {
        Dog d = new Dog();
        d.sound();
        d.sleep();
    }
}
```

6. Types of Abstraction

Type	Meaning
Partial Abstraction	Some methods implemented
Full Abstraction	All methods abstract (mostly using interfaces)

7. Use Cases of Abstract Classes

- Game character systems
- Banking software
- Employee management systems
- Vehicle simulation programs

8. Abstract Class vs Interface

Feature	Abstract Class	Interface
Methods	Abstract + Concrete	Mostly Abstract
Variables	Normal variables allowed	Only constants
Constructors	Yes	No
Inheritance	Single	Multiple

9. What is an Adapter Class?

An Adapter Class in Java is a helper class that provides empty implementations of interface methods. Instead of implementing all methods manually, programmers extend the adapter class and override only required methods.

Important: Adapter classes are commonly used with event handling in Java GUI programming.

10. Why Use Adapter Classes?

- Reduce unnecessary code
- Avoid implementing every method
- Simplify event handling
- Improve readability

11. How Adapter Classes Work

Suppose an interface has 7 methods, but you only need 1. Without adapter class, you must implement all 7 methods. With adapter class, you override only the needed method.

12. Adapter Class Example

```
import java.awt.event.*;

class MyWindow extends WindowAdapter {

    public void windowClosing(WindowEvent e) {
        System.out.println("Window Closed");
    }
}
```

13. Common Adapter Classes

Adapter Class	Used For
WindowAdapter	Window events
MouseAdapter	Mouse events
KeyAdapter	Keyboard events
FocusAdapter	Focus events

14. Real World Use Cases

- Desktop applications
- Java Swing applications
- GUI event handling
- Large enterprise software

15. Advantages and Disadvantages

Advantages:

- Cleaner code
- Easy maintenance
- Improves abstraction
- Reduces duplicate code

Disadvantages:

- Complexity for beginners
- Extra inheritance layers
- Can increase hierarchy complexity

16. Last Minute Exam Notes

- Abstract class uses abstract keyword.
- Cannot create object of abstract class.
- Child class must implement abstract methods.
- Adapter classes provide empty method bodies.
- Adapter classes are mostly used in event handling.